

Business Case: Connected Factory Transformation

Case overview

Company: NovaTech Manufacturing, a fictional ₹2,000 crore manufacturer with three plants, eight warehouses and 5,000 SKUs.

Business challenge: The company has installed some sensors, dashboards and ERP capabilities, yet unplanned stoppages still disrupt production. Plants use different equipment and product identifiers, production teams often update planners through spreadsheets, and customer delivery commitments are revised late.

Proposed investment: Build a connected operations capability around a critical production line, prove its value, then scale it across plants. The solution combines equipment data, SCADA/DCS where applicable, MES, ERP, process mining, workflow automation, analytics and a limited digital twin. Change management and cybersecurity are part of the delivery plan.

1. Problem statement

NovaTech faces three linked problems:

1. **Production interruptions are detected locally but communicated slowly.** An operator sees a machine alarm, while the MES work order and ERP delivery plan may remain unchanged until someone updates them.
2. **Maintenance response varies by shift and plant.** Teams receive alerts, but ownership, inspection priority and escalation rules are inconsistent.
3. **Managers cannot compare the same problem across plants reliably.** Equipment IDs, downtime reasons, timestamps and product codes differ between systems.

The result is an operational decision gap: the company may know that a machine stopped without knowing promptly **which order is affected, what recovery options exist, who must act, and whether a customer commitment is at risk.**

Illustrative baseline to validate

Measure	Current baseline	Why it matters
Unplanned downtime on selected assets	1,200 hours/year	Lost productive capacity
Financial impact of downtime	₹50,000/hour	Contribution at risk, subject to finance validation
Annual scrap and rework cost	₹4 crore	Quality loss
Inventory	₹100 crore	Working capital
OTIF	89%	Customer service
Time to communicate a material production delay to planning	To be measured	Speed of response

The project team must verify these figures, especially whether a downtime hour truly causes lost contribution or whether production is recovered later.

2. Objective and scope

Business objective: Improve production reliability and delivery decisions on a critical line, then repeat the proven operating model across suitable sites.

Initial pilot: One production line in Plant A, its maintenance workflow, the affected MES work orders and their linked ERP customer orders.

Proposed success criteria:

- A stoppage is associated with the correct asset, production work order and customer order.
- Every actionable maintenance alert has an owner and recorded disposition.
- Planners can assess delivery impact using current production status.
- Downtime and quality improvements are measured against an agreed baseline.
- Operators and maintenance teams use the process routinely, including during exceptions.

A project is successful only when people make better decisions and operating results improve—not when a dashboard merely goes live.

3. How to approach the transformation

Phase 1 — Diagnose and baseline: months 0–3

Map the real process. Use MES, maintenance and ERP event logs to trace a stoppage from alarm to inspection, repair, production update and planning decision. Process mining can reveal waiting time, repeated approvals, manual handoffs and rework.

Check data readiness. Match equipment IDs across sensor and maintenance records; link MES production work orders to ERP customer orders; agree on downtime reason codes, timestamp rules and KPI definitions. Different codes can remain in each system if a governed mapping reconciles them.

Select the pilot. Prioritize an asset with material downtime, reliable enough data, a practical response action and an accountable plant owner.

Decision gate: Is the problem valuable and measurable enough to justify a pilot?

Phase 2 — Pilot and validate: months 3–12

Connect the selected line's equipment events to the operational systems. Define who receives an alert, when inspection is required, how exceptions are escalated and when the planner must review affected orders.

Introduce analytics only for a decision the team can act on—for example, flagging a pattern of repeated motor vibration. Build a digital twin for a bounded question such as: *What happens to output and delivery dates if this machine is unavailable for six hours?*

Compare pilot outcomes with the baseline. Review false alerts, missed events, response time and the workload created for operators.

Decision gate: Did the pilot produce a credible operational and financial improvement that can be repeated?

Phase 3 — Scale and optimize: months 12–24

Standardize identifiers, interfaces, training and operating rules. Extend the solution to suitable lines and plants while allowing for genuine equipment and process differences. Review benefits by site rather than assuming the pilot result will transfer unchanged.

Decision gate: Are improvements sustained across sites after implementation support is reduced?

4. End-to-end solution

A. Connected operational data

A motor on Line 2 stops at 10:15. The equipment event records **asset M-12** and its timestamp. SCADA displays the alarm to the operator; a DCS, where relevant to that process, maintains its defined control functions. MES records the interruption against **production work order WO-458**. Integration maps that work order to **ERP sales order SO-927**.

The planner can now see that the production interruption may affect a promised delivery. The planner reviews options and the customer commitment under the company's approval rules. The system supplies a connected view; it does not automatically change the promise.

B. Maintenance workflow

For an actionable alert:

1. Assign a named maintenance owner.
2. Check asset criticality and the operating condition.
3. Inspect or schedule an inspection under an agreed response rule.
4. Record the finding, action and time taken.
5. Escalate exceptions—for example, unavailable parts or a safety concern.
6. Close the alert only after the action and outcome are documented.

Workflow automation can route tasks, send reminders and track overdue work. Technical and safety decisions remain with qualified people.

C. Analytics and digital twin

Analytics identifies repeated stoppages, unusual conditions and differences between comparable assets. The digital twin uses a validated representation of the line to compare recovery options.

For example, under **illustrative assumptions**, a line producing 100 units/hour loses six hours of operation: **600 units of planned output are at risk**. If another line can make 60 units/hour for those six hours, it produces **360 units**, leaving a **240-unit output gap**. If 100 finished units can serve the affected order, the potential **customer-order shortfall is 140 units**. Stock covers demand; it does not restore lost production output.

Before acting, managers must verify repair duration, alternative-line capacity, product compatibility, inventory availability and dispatch deadlines.

D. Cybersecurity and resilience

Inventory connected devices and interfaces; restrict access by role; separate critical operational networks where appropriate; plan controlled patching, backups and restoration. Test what the plant can safely observe and operate if a monitoring or integration service is unavailable for one shift. The incident procedure must specify when to pause operations or use an approved manual process.

E. People and adoption

Operators help define useful alerts. Maintenance leads own triage and inspection priorities. Planners review order impact. Plant managers review overdue actions, false alarms and operating results.

Training uses actual tasks: a valid alert, a false alert, an unavailable technician and a machine stoppage during a priority order. Adoption measures include alert acknowledgement, assigned ownership, completed inspections and use of the connected production view—not just login counts.

5. Illustrative investment and return

The following is a **proposed financial model**, contingent on baseline validation and a successful pilot.

Annual benefit after scale	Calculation	Estimated value
Downtime reduction	300 hours × ₹50,000 validated contribution/hour	₹1.50 crore
Reduced scrap and rework	₹4.00 crore baseline reduced by 20%	₹0.80 crore
Inventory carrying-cost reduction	₹8 crore less inventory × 10% annual carrying cost	₹0.80 crore
Less administrative rework	Validated process-time and error savings	₹0.40 crore
Total annual benefit		₹3.50 crore

Investment assumptions: ₹4.00 crore initial implementation and ₹0.80 crore annual running cost.

- **Annual net benefit after running cost:** ₹3.50 crore – ₹0.80 crore = **₹2.70 crore**
- **Simple payback after full-scale benefits begin:** ₹4.00 crore ÷ ₹2.70 crore ≈ **1.5 years**
- **Three-year undiscounted net benefit:** 3 × ₹2.70 crore – ₹4.00 crore = **₹4.10 crore**

This payback is a *steady-state illustration*. A real investment case must phase costs and benefits by month because the pilot and rollout take time. It must also avoid counting the same recovered production twice as both downtime savings and increased sales. The ₹8 crore inventory reduction releases working capital; the model counts only its **₹0.80 crore annual carrying-cost benefit** as recurring value.

6. What results should be reported?

Use three levels of evidence:

Level	Example measures	What it establishes
Implementation	Assets connected; event mappings validated; users trained	The capability exists
Adoption and decisions	Alerts acted on; response time; planner reviews; exceptions resolved	People use it in daily work
Business outcomes	Downtime, scrap, inventory carrying cost, OTIF and validated financial benefit	The operation improved

Do not present the financial estimates as achieved results. After the pilot, report the actual baseline period, comparison period, production mix, unusual events and any other changes that could explain the difference.

Executive recommendation

Approve the diagnostic and bounded pilot, with funding for scale released at the validation gate. Assign a plant operations sponsor, maintenance owner, production planning owner, IT/OT integration lead, cybersecurity lead and finance partner. Require them to demonstrate a trusted event-to-order link, routine use of the response workflow and a finance-validated improvement before expanding to other plants.

Class discussion: *If the pilot reduces downtime but creates too many false alerts and extra work for technicians, should NovaTech scale, revise or stop? What evidence would support that decision?*

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